



Brief description

The function block controls a "continuously determined force with a defined force ramp in a continuous motion". Depending on the scaling, the function block is suited for power determination and torque determination (positive power/torque determination in the sense of positive velocity determination). To end the motion, the function block instance has to be interrupted by another function block instance which issues a new command.



If this function block requires power added before via an instance of the [MC Power](#) (for real axes).



Depending on axis configuration and interpolation in the drive/control, MC_TorqueControl responses as follows:
The torque ramp in the drive is calculated in case of **interpolation in the drive**. MC_TorqueControl writes to the parameters:

- Torque input has an effect on [S-0-0080](#), Torque, force command value
- TorqueRamp input has an effect on [S-0-0822](#), Torque, force ramp
- [S-0-0823](#), Torque/force ramp time The control firmware configures the parameter with a fixed value of 1000ms

The torque ramp is calculated in the control in case of **interpolation in control**.

- The control writes cyclically on "[S-0-0080](#), Torque, force command value".
- The generation of a torque ramp in the drive is disabled ([S-0-0822](#) = 0, [S-0-0823](#) = 0).
- The TorqueRamp input corresponds to a change in torque within 1000ms.



The functionality is supported by the following axis types:

- IndraDrive with interpolation in the drive
- IndraDrive with interpolation in the control
- SercosDrive
- Controller axis

The "InTorque" signal is reset if the function block instance is interrupted by another function block instance (CommandAborted) or if the edge of its "Execute" input is falling.

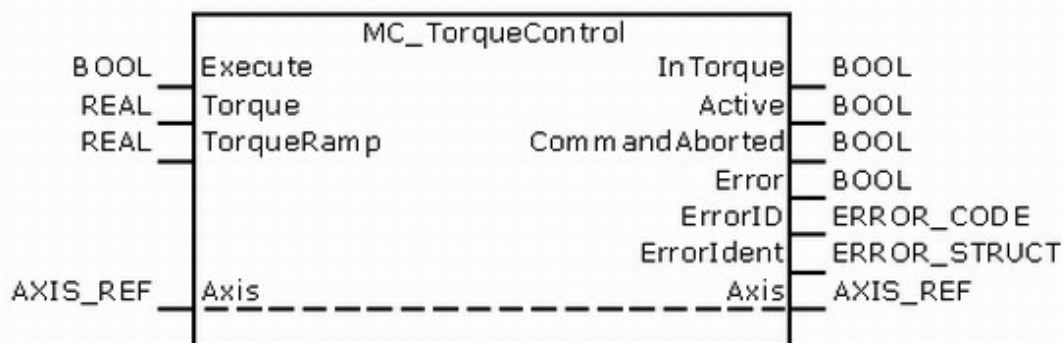


If "Execute" is disabled, the drive keeps running on the same velocity.

Target system	Library/folder
IndraLogic XLC/IndraMotion MLC 14VRS MOT function package	ML_PLCopen / Motion

Library assignment

Interface description



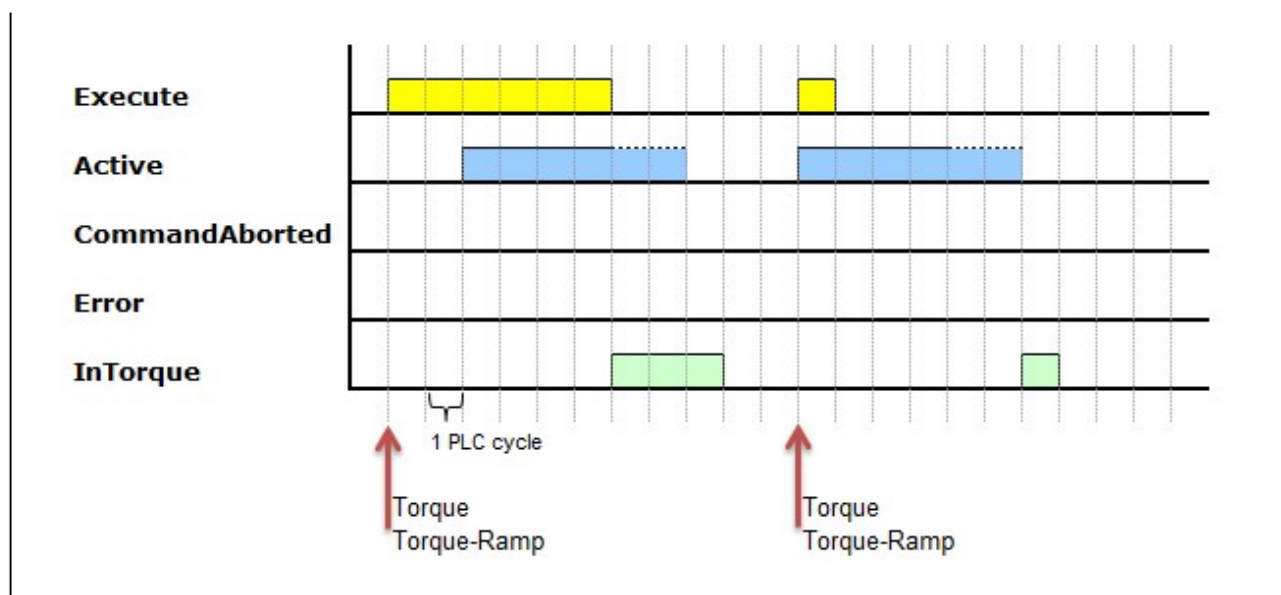
MC_TorqueControl function block

	Name	Type	Comment ¹⁾
VAR_INPUT	Execute	BOOL	Starts the motion at a rising edge. The input values are applied at each edge.
	Torque	REAL	Maximum value of the power/torque (always positive, value does not necessarily have to be reached).
	TorqueRamp	REAL	Power/torque difference which can be reached within the ramp time
VAR_OUTPUT	InTorque	BOOL	Power/torque reached (for the first time)
	Active	BOOL	Data processing runs when preprocessing is complete
	CommandAborted	BOOL	Command aborted by subsequent command
	Error	BOOL	Indicates that an error occurred in the function block instance
	ErrorID	ERROR_CODE	Brief note on error cause
	ErrorIdent	ERROR_STRUCT	Detailed error information
VAR_IN_OUT CONSTANT	Axis	AXIS_REF	Contains information on the specific axis

Interface of the MC_TorqueControl function block

Signal-time diagram

Edge-controlled function block with permanent processing - processing completed



Signal course of time of the MC_TorqueControl

Activation/deactivation of "Torque"

Applying the torque command value to the master data telegram has to be enabled in the parameter "[A-0-0007](#), Axis configuration", bit 2.

Parameters used (Sercos-based axes)

The function block operates together with the following Sercos parameters:

- [S-0-0091](#), Bipolar velocity limit value (to be set before the function block call)
- [S-0-0823](#), Ramp time (to be set before function block call)



Set ramp time to 1,000 ms and specify the torque/power difference to be programmed related to one second at the function block input.

- [S-0-0822](#), Torque/force ramp, applied by the function block.



The parameters [S-0-0822](#) and [S-0-0823](#) are supported by the drive variants from HNC3x Sercos and MPH04.

Troubleshooting:

The function block MC_TorqueControl generates the following error messages:

- [MLC_TABLE](#) - Error when accessing the axis/control.

Footnotes

- 1) The measuring units of time, position, velocity, acceleration, deceleration and power/torque depend on the effective scaling.

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